

Advanced simulation and optimisation of the telecommunications network

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This paper describes current research which supports improved planning and operation of the telecommunications network. A future holistic approach to network simulation is envisioned which would include the interactions of network infrastructure, operating processes and software systems with investment and revenues. Near-term approaches to simulating and optimising network designs are reviewed and medium-term objectives are defined.

1. Introduction

In line with the rapid growth of telecommunications networks in recent years there has been a corresponding increase in the level of network complexity. As a consequence, it is now generally accepted that advanced computer aided simulation and analysis methods are essential aids to the management of large telecommunications networks. It is desirable that such simulation tools are integrated into a coherent system. This paper describes possible research initiatives to support the simulation of management processes and also to optimise their interactions. Figure 1 shows the key interactions between the different management functions.

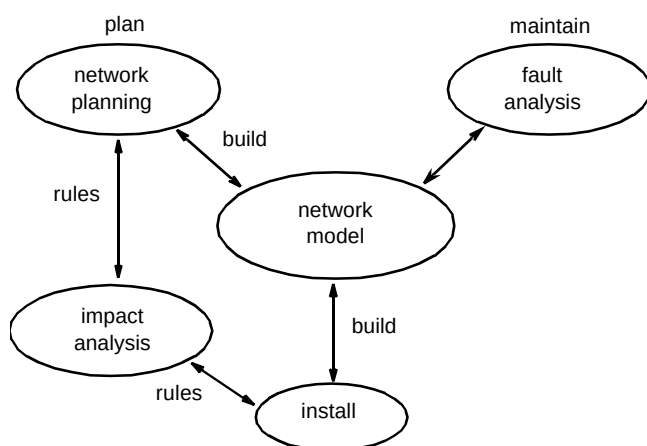


Fig 1 Interaction between management functions.

The 1990s will be recalled as the decade of business process re-engineering. Most large, conventional manufacturing engineering organisations have already applied some re-engineering to improve their overall process efficiency

resulting in the establishment of streamlined management structures, optimised budgets and reduced staffing levels.

In the future, information will assume a much greater level of importance. It is becoming increasingly evident that many businesses will achieve and actively maintain their competitive edge by continuously gathering, analysing and exploiting operational data in new and novel ways. Large-scale system models will be analysed and new near-optimal structures will be evolved. Macro-level system analysis and optimisation can now be attempted because of recent developments in mathematical methods, artificial intelligence, and low-cost, high-performance computers. These computers can solve complex problems by processing vast amounts of data, using new methods, providing better solutions and giving deeper insight into the complex structure of network problems. The primary goals for telecommunications network operators are to develop methodologies for:

- planning — optimum network planning methodologies are required to minimise the cost of building new network structures,
- fault analysis and maintenance — accurate identification and repair of faults with the minimum disruption to network operation,
- business modelling — evaluation of the effects that selected, individual or grouped changes will ultimately have upon the whole business and organisation,

- impact analysis — determination of effective network provisioning policies to aid the design of new components and accurately predict the impact of next-generation equipment on the network and business.

2. Simulating network process

This section considers the general requirements for process modelling, service provisioning policy, system reliability, and new technology impact analysis.

2.1 Process modelling

An envisaged business model will assess the impact of management policies upon organisational structures, in particular network processes and resource levels (see Fig 2). This model will be built from a number of separate communicating processes, as described in section 4. Strategically, there are major advantages associated with the creation of an overall business model as service provisioning policy, system reliability and new technology impact can all be considered together.

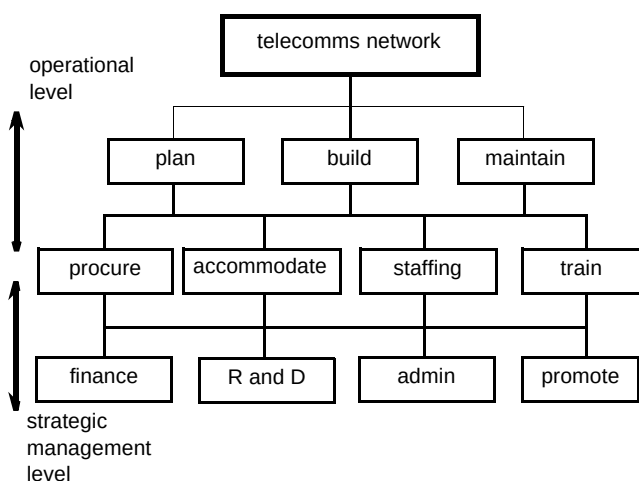


Fig 2 Plan/build/maintain management processes.

2.2 Service provisioning policy

Proactive and reactive service provision can be defined and compared as follows.

- Proactive — where demand is anticipated and spare service capacity is installed in the network before an order is received from a customer. The advantage is that the service can be provided for the customer with the minimum of delay. Unfortunately, this requires capital expenditure and installation costs for equipment

some of which may not be required. Clearly, very accurate forecasts of future demand are essential using this approach.

- Reactive — where demand is satisfied only after an order is received. The tools described in this paper allow design plans to be produced more rapidly with component inventories automatically created and orders rapidly processed using electronic interfaces with suppliers and contractors. As a result of these improvements, response times will be reduced. Other advantages include reduced stock inventory levels, overheads and capital expenditure.

2.3 System reliability

Telecommunications network simulation and analysis can be used to resolve the following issues:

- viability of a policy of immediate fault repair as opposed to service restoration by use of built-in spare capacity,
- the optimisation of lifetime costs by effectively balancing capital investment and maintenance costs,
- the effect of repair activities upon network reliability,
- the introduction of resilient, fault-tolerant systems into the network (intelligent redundant stand-by modules),
- the relationship between system reliability and maintenance staffing levels.

It is anticipated that the proactive provisioning policy will be applied more often in the future. Business processes and staff structures will therefore have to be adapted for these new methods of working. The introduction of the overall business model, incorporating the planning and maintenance tools described in this paper, could provide economic justification for the adoption of new management policies. Introduction of the integrated system described should generate significant savings in capital expenditure, installation and maintenance costs.

2.4 New technology impact analysis

The impact of introducing new technology can be assessed, as next-generation network components can be specified in software, and design plans created to determine their effect on network topology and reliability. Impact analysis is even possible before detailed component design and development commences.

The impact of new technologies on telecommunications network build and provisioning costs is difficult to assess. Simulation strategies, such as sensitivity analysis in

conjunction with planning software, are therefore being investigated to ensure that the overall implications of these business decisions are more clearly understood. Simulation will be undertaken using Monte Carlo methods to determine the effects that parameter variation will have upon the cost of building and providing telecommunications network services. In this system, the following temporal and strategic factors must be considered when determining provisioning policies:

- variation in demand, equipment and labour costs with time,
- impact of new technologies on cost and quality of service,
- impact of operational processes on network costs and performance,
- effects of process interaction.

3. Infrastructure data models and data capture

To understand the real telecommunications network it is important that a comprehensive data model is developed which clearly represents the relationships between the physical and abstract entities which exist in actual networks. At the detailed level, this could describe both the concrete physical association between a cable and its duct and the abstract association of a time slot within a transmission frame. A simplified database schema for planning is shown in Fig 3. This model employs the object modelling technique (OMT) notation (see Paul et al [1] for a description of inheritance, association and aggregation). In general, the telecommunications network can be represented by a set of nodes connected by sets of links:

- nodes hold information relating to equipment enclosures and the number of installed network components,
- links store information relating to the duct layout and cable connectivity.

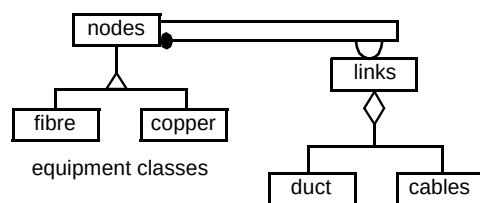


Fig 3 General telecommunications network data model.

A data evolution strategy is being developed to populate the main data model in a cost-effective manner. In many instances, data is presently held in paper records. A number of data-capture factors and options must be considered.

An option is to capture paper-based records in a faster format and convert this information into the equivalent vector and alpha-numeric format as required. Unfortunately, faster graphics data cannot be easily read by computer programs and therefore, in this case, manual data manipulation is often required.

A second option, although more costly to implement, is to convert all existing paper-based records into the vector and alpha-numeric format. An advantage of this approach is that vector and alpha-numeric data can be directly read by computer programs.

At present, a comprehensive digital database for external plant and equipment is being built for the BT UK telecommunications network. New data-capture systems are currently being installed to gather network data from paper records. All of the computer-based management functions briefly outlined in the following sections rely upon those databases which store information relating to network structures and components.

Databases are being designed with features that ensure data integrity through continuous validation.

A number of key issues emerge and these are listed below:

- implementation of version control and management systems,
- management of data structures to eliminate invalid data,
- data validation by visualisation and inspection,
- data security,
- use of multimedia to capture non-numeric data, such as pictures of typical equipment and specific locations,
- node and link capacities.

A strategy is required to manage multiple access to network data where several planners may require simultaneous access to a single data store. A networked database system and wide area network (WAN) will be implemented so that data can be easily accessed by many

users from numerous different locations. A typical arrangement is shown in Fig 4.

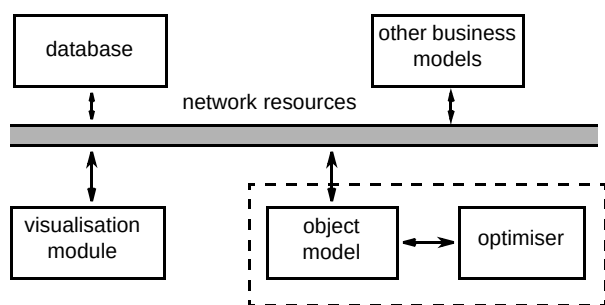


Fig 4 A typical network resources arrangement.

4. Automated methods

Assuming the existence of a complete and accurate electronic record-base for network infrastructure, optimisation techniques for network planning can now be considered, together with some advanced network management possibilities such as the use of artificial neural networks for fault analysis.

4.1 Optimisation techniques

At present, network planning usually involves human skills and intelligence, together with calculators and computer spreadsheets. Research has been undertaken in order to improve network planning methods so that computer-based optimisation strategies can be applied to determine near-optimal structures, both for standard telephone systems [2–4] and for passive optical networks (PONs) [5].

The abundance of difficult optimisation problems encountered in practical settings, for example in telecommunications, logistics, financial planning, transportation and production, has motivated a proliferation of optimisation techniques. The following three methods have been identified by the Committee on the Next Decade of Operations Research [6] as ‘extremely promising’ for future treatment of practical applications.

- Generic methods seek to imitate the biological phenomenon of evolutionary reproduction [5, 7]. The principles of evolutionary computing have been applied to solve the network planning problem. Evolutionary computing techniques are based upon Darwin’s theory of evolution where a population of individuals, in this case potential solutions, compete with one another over successive generations —

‘survival of the fittest’. After a number of generations, the best solutions survive and the less fit are gradually eliminated from the population.

- Simulated annealing is based upon a physical process in metallurgy [8, 13]. In simulating the cooling process, a sequence of states is generated typically through random selection method. In this approach, a ‘temperature-dependent’ noise term in the algorithm promotes transitions that increase the probability of finding a near-optimal global solution. Research is currently being carried out to assess the capability of this method to solve a version of the network planning problem, taking into account component positioning, cable routing and duct layout.
- Tabu search may be regarded as a technique based on selected concepts from artificial intelligence [9, 10]. It is a general heuristic procedure for guiding a search to obtain good solutions in a complex space. This method employs a memory function which improves search efficiency by reducing the probability of revisiting invalid solution areas. A tabu search algorithm is currently being developed to solve the network planning problem. The results obtained from this research project will be reported in a later paper.

All of the above methods are currently being evaluated with regard to relative complexity, ease of implementation, search efficiency, hybridisation and potential capability for application within a parallel-processing framework. Ongoing research will compare the relative performance of these methods for network planning.

4.2 Artificial neural networks

Network faults should be analysed, located and repaired with the minimum of disruption to system operation. In the future, it will become even more important to minimise loss of service through component failure because a single multiplexed component will handle many separate data channels. At present, fault location is normally the domain of a human expert, since there are few computer systems currently in use which can identify the cause of complex faults. Consequently, alternative intelligent fault-location methods are now being evaluated [11].

Many researchers believe that neural networks offer the most promising unified approach to building truly intelligent computer systems [12, 13]. A simple 324 feed-forward neural network is shown in Fig 5, which illustrates a three-layer structure, the neurons, their interconnections and weights. Artificial neural networks are greatly simplified models of the central nervous system and are networks of highly interconnected neural computing elements which have the ability to respond to input stimuli and to learn to adapt to their environment. Neural networks

employ parallel distributed processing architectures [8]. Hammerstrom [12] clearly describes the three major advantages of neural networks.

- Neural networks are adaptive — they can take data and learn from it. This ability differs radically from standard software because it does not depend upon the prior knowledge of rules. In addition, neural networks can reduce development time by learning underlying relationships even when they are difficult to find and describe. They can also solve problems that lack existing solutions.
- Neural networks can generalise — they can correctly process information that only broadly resembles the original training data set. Similarly, they can handle imperfect or incomplete data, providing a measure of fault tolerance. Generalisation is useful in practical applications, such as fault analysis, because, in the real world, data is often noisy.
- Neural networks are nonlinear — they can capture complex interactions among the input variables in a system.

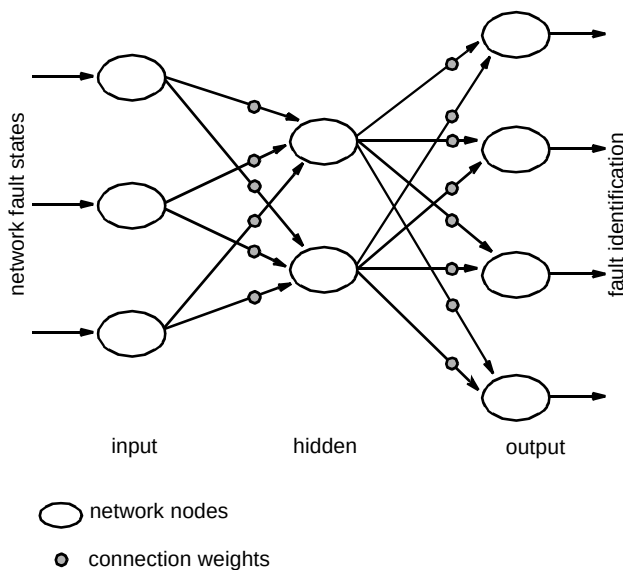


Fig 5 An artificial neural network.

Preliminary research is being undertaken to determine the viability of neural networks, operating in conjunction with an expert system, to identify the nature of difficult network faults where complex interacting relationships exist between the different fault parameters. In this context, neural networks are being developed to analyse data in order to achieve improved decision making for maintenance purposes. Neural networks have a number of important features with regard to their application to network fault analysis, namely, they can learn to recognise subtle

relationships 'buried' within training data and are able to adapt to widely differing telecommunications network configurations.

Preliminary results indicate that further research is required to improve training efficiency. It is anticipated that artificial neural network fault-location systems will be able to accurately identify the location of faults within a network and, in conjunction with an expert system, determine the best course of action for maintenance and repair. Furthermore, it is also expected that neural network fault-diagnosis systems will ultimately lead to improved fault analysis, reduced maintenance costs, fewer network interventions for repair, more rapid repair of faulty components and restoration of service.

5. The way forward

In conclusion, near-term goals, medium-term goals and some strategic factors in the development of advanced network simulation tools are now considered.

5.1 Near-term goals

The near-term goal for the research reported in this paper is the development of a suite of stand-alone strategic computer-aided network-simulation and decision-support tools to improve the operation of the telecommunications network. Key areas for consideration are reviewed below.

- Infrastructure data model

A large-scale model of the telephone network is in the final stages of design, and alternative cost-effective methods of data capture are now being investigated. It is recognised that WANs are an essential technology to be used within the business process model. A strategy for the implementation of the WAN will be drawn up in the near future.

- Planning

Prototype auto-planning tools for specialised networks have been designed and developed and are now being evaluated. It is envisaged that an automated planning system could be implemented which incorporates the best features of the prototype systems. To prove the acceptability of this system, experienced planners will be given a version of the planning tool to test and evaluate, prior to validation and full-scale application.

- Maintenance, fault analysis

Data relating to network performance is being gathered to compile a comprehensive library of 'fault states'. This will enable a detailed programme of work for the development of a fault-detection system based upon neural network technology.

At present, management relies largely upon the consensus views of engineers with considerable local experience of network design and maintenance. The development of an intelligent-simulation and computer-aided support system will enable strategic and detailed network planning and maintenance policies to be speedily evaluated prior to more rapid implementation. Strategic policies will be consistently applied to produce near optimal network solutions across the whole organisation. In the future, these tools will reduce the workload for the user because the tedious burden of detailed calculations will be carried out by the computer. In the near term, stand-alone tools to plan, build and maintain the telecommunications network will be developed and a strategy will be created for the integration of these tools into an overall business model.

5.2 Medium-term goals

In the medium term, the tools described above, and in Paul and Tindle [5], will be integrated into an overall business model. Strategically, this holistic approach to network modelling and simulation has a number of advantages, for example, during the planning process many factors such as service provisioning policy, system reliability and new technology can all be taken into account.

Clearly, further significant developments in the areas of artificial intelligence, neural networks and optimisation techniques will be incorporated into the simulation and decision-support tools. It is anticipated that the developments outlined in this paper will ultimately lead to greatly improved operation of the telecommunications network.

5.3 Strategic issues

The underlying objective of UK Government policy is to provide customers with the widest range of telecommunications services at competitive prices. Current Government policy has concentrated on creating a framework for infrastructure development and investment [14, 15]. As the process of European liberalisation continues, telecommunications companies are well placed to export a wide range of products and services to world-wide markets. These services will include telecommunications software for the overall operation of the network using advanced methods such as those described in this paper. This liberalisation has resulted in higher levels of competition in the UK telecommunications sector, making it essential that telcos have the necessary analytical, financial and customer-oriented skills to meet the challenge of providing a wide range of reliable, low-cost services.

In the telecommunications sector, sustainable competitive advantage can be achieved by applying new

advanced 'intelligent' technologies which empower managers, enabling them to analyse complex systems, determine appropriate management strategies and expedite projects and business cases. Implementation of the methods described above will provide managers with improved insight into network operational issues ensuring that the telecommunications sector retains and enhances its reputation for the excellent management of complexity and change.

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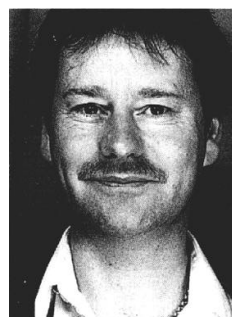
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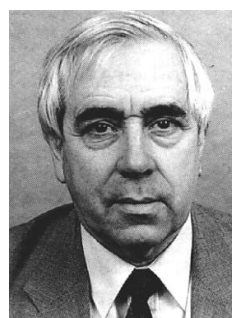
A number of industrially sponsored projects are currently being undertaken including engineering software development for large engineering networks, evolutionary computing, intelligent system development, parallel computing, artificial neural networks for fault state analysis, networking and image analysis.

Before joining SEAT he was employed by Internal Research and Development IRD, part of Rolls Royce Group, as leader of the Electronics and Control Group. An active member of the IEE, he has held the posts of Chairman, Secretary and Treasurer of the Electronics, Computing and Control Section of the IEE North East Centre. At present he is also a Member of IEE Professional Group, Information Systems and Networks.



Steve Brewis graduated from Sunderland University in 1980 with a Degree in Mechanical and Electrical Engineering. He continued his studies and gained an MSc in Management from the University of Northumbria. He is currently employed as a network systems development manager with BT where he is actively involved in developing novel ways of managing the telecommunications network. He has been employed as a part-time lecturer at the University of Northumbria in Energy Management and in 1985 he was awarded the Gas Energy Management Award by British

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Hugh Ryan graduated from the University of Strathclyde in 1960 with a first class honours degree in Electrical Engineering. He was subsequently awarded a PhD by the University of Durham in 1965. He worked with Reyrolle Switchgear from 1960 to 1985, latterly as Head of Research, Reyrolle Technology. He joined the University of Sunderland in 1985 initially as Head of Department and Professor of Electrical and Electronic Engineering. He is currently Director of Research in the School of Engineering and Advanced Technology. His research interests are wide ranging and he is

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